BPJS Kesehatan
Badan Penyelenggara Jaminan Sosial

HEALTHKATHON 2026
DETECT SMARTER, PROTECT JKN

Smarter Protection for a Stronger JKN: Efisiensi Risiko Program JKN

Inisiatif Untuk Efisiensi

HUMAN-CENTRED AI FOR HEALTH FUTURE

From Problem to Prototype: Building AI Solution - Basic Level

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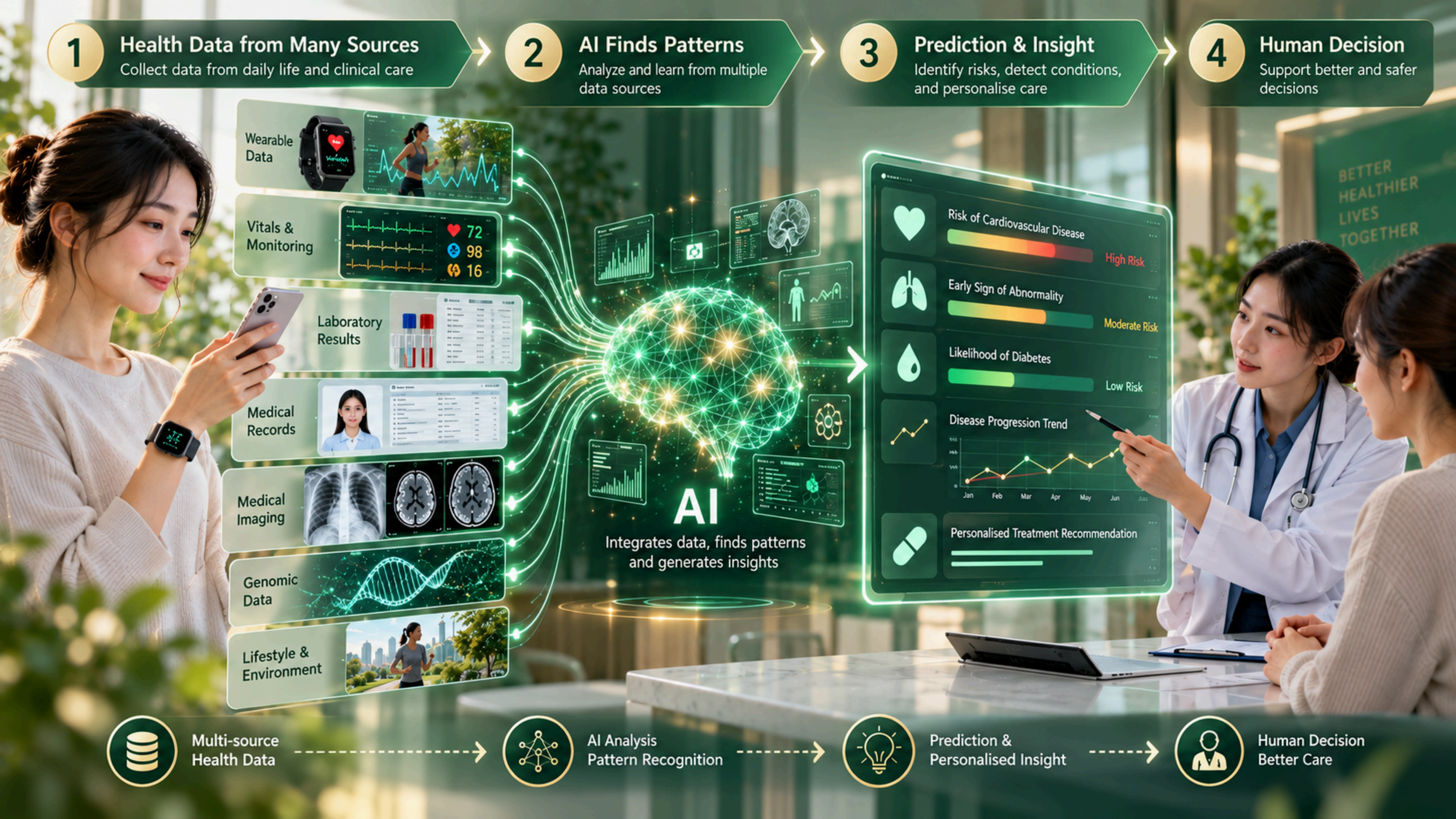

1
Health Data from Many Sources
Collect data from daily life and clinical care
2
AI Finds Patterns
Analyze and learn from multiple data sources
3
Prediction & Insight
Identify risks, detect conditions, and personalise care
4
Human Decision
Support better and safer decisions
Wearable Data
Vitals & Monitoring
72
98
16
Laboratory Results
Medical Records
Medical Imaging
Genomic Data
Lifestyle & Environment
AI
Integrates data, finds patterns and generates insights
Risk of Cardiovascular Disease
High Risk
Early Sign of Abnormality
Moderate Risk
Likelihood of Diabetes
Low Risk
Disease Progression Trend
Jan
Feb
Mar
Apr
May
Jun
Personalised Treatment Recommendation
BETTER HEALTHIER LIVES TOGETHER
Multi-source Health Data
AI Analysis Pattern Recognition
Prediction & Personalised Insight
Human Decision Better Care

The Fraud Ecosystem: Mapping the Vectors of Exploitation

The Primary Modus Operandi Matrix

Fraud Type	Execution Method	Digital Trace / Audit Flag	Systemic Impact
Phantom Billing	Submitting claims for surgeries or treatments (like cataracts/ physiotherapy) that never occurred.	Patient absence on specific dates; zero corresponding pharmacy records.	Massive, direct financial extraction; high severity.
Upcoding	Falsifying ICD-10/ICD-9-CM codes to a higher severity level to trigger higher INA-CBG tariffs.	Statistical anomalies in disease frequency compared to regional averages.	Overpayment per episode; clinically complex to prove.
Service Unbundling	Splitting a single care episode into multiple distinct claims to inflate total reimbursement.	Multiple claims filed for the same patient within a suspiciously short timeframe.	Artificial inflation of claim volumes and administrative bloat.
Illegal Billing	Illegally charging fully-covered JKN patients out-of-pocket for mandatory services.	Patient whistleblower reports; unrecorded off-book hospital revenues.	Direct financial harm to citizens; destroys trust in Universal Health Coverage.

TYPES OF MACHINE LEARNING

Three Approaches to Teaching Machines

expert guidance (teacher)

labeled data (validated records)

SUPERVISED LEARNING

Learns from labeled examples, where input data is paired with corresponding output labels.

Classification

(e.g., Disease detection, Claim approval categorization, Image diagnosis)

Regression

(e.g., Risk prediction, Cost forecasting, Patient outcome prediction)

clustering (grouping similar data)

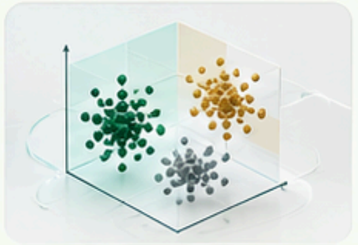
pattern discovery (hidden insights)

UNSUPERVISED LEARNING

Finds hidden patterns and structures in unlabeled data without human intervention.

Clustering

(e.g., Patient segmentation, Customer grouping, Anomaly detection)



Dimensionality Reduction

(e.g., Data visualization, Feature extraction, Simplification of medical or insurance data)

robot/agent (intelligent decision maker)

reward (better outcomes)

REINFORCEMENT LEARNING

Learns through trial and error interactions with an environment to maximize cumulative rewards.

Healthcare AI

(e.g., Treatment recommendation support, Workflow optimization, Scheduling)

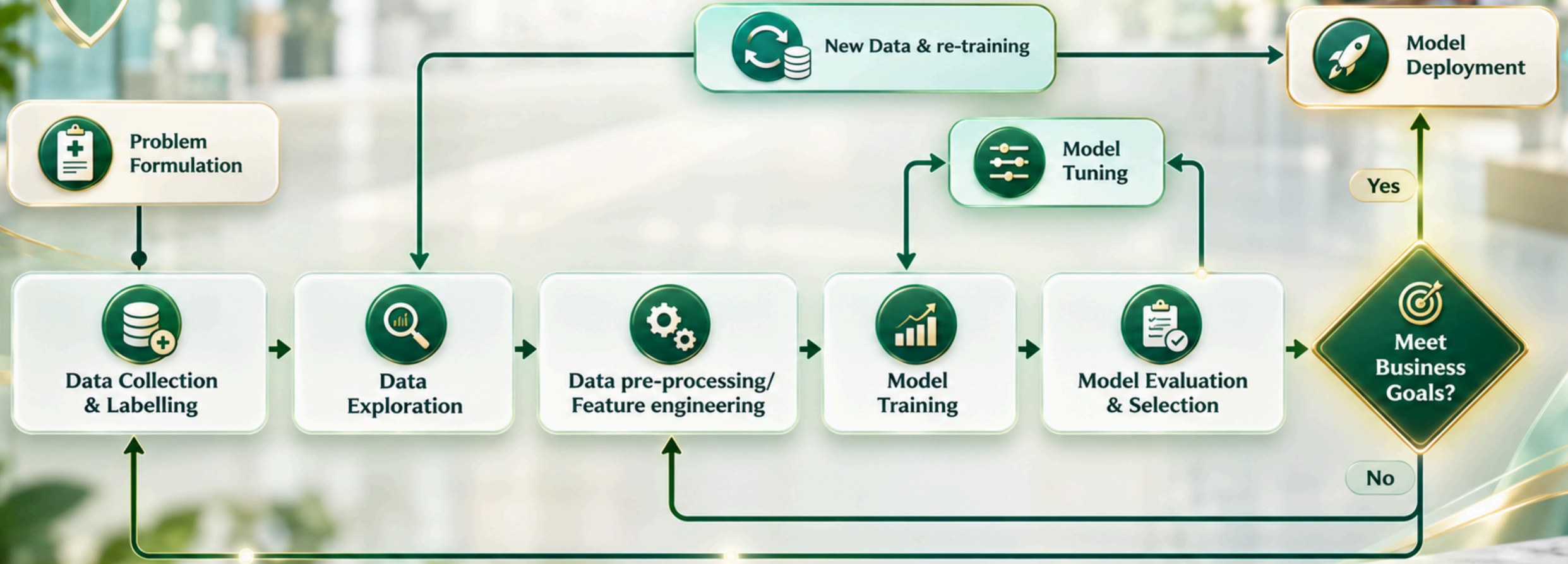
Robotics / Automation

(e.g., Autonomous navigation, Hospital logistics, Claims process automation)

1
Predictive AI
Predicts what might be happening
2
Generative AI
Explains, summarises and answers
3
Agentic AI
Takes actions for you
Health Data
Cough Audio
Vitals & Monitoring
72
HR
98
SpO2
16
RR
Spectrogram
Medical Records
AI Model
CNN
Prediction
Likely Normal Cold (Common Flu)
82%
Possible
12%
Others
6%
Summary
Your cough is most likely a common viral respiratory infection.
Usually improves within 1–2 weeks.
Rest, hydration and symptom relief are recommended.
Key Points
No urgent signs detected.
Monitor if fever persists or breathing worsens.
Consider a doctor visit if symptoms last > 2 weeks.
Questions to Ask Your Doctor
Do I need a chest X-ray?
Is a flu test or COVID-19 test necessary?
What medication is suitable for me?
Are there any red flags I should watch for?
Book Appointment
Appointment Confirmed
Mon, 14 Apr 2026
10:00 AM
Order Medicine
Medicine Ordered
Schedule Lab Test
Lab Test Scheduled
Care Plan
Doctor Appointment
Mon, 10.00 AM
Medicine Delivery
Arriving Tomorrow
Lab Test
Tue, 8.00 AM
Reminders
Take medicine as prescribed
Monitor your symptoms
Rest and stay hydrated

Machine Learning Pipeline Framework

This flowchart outlines the process of building and deploying an ML model, from problem formulation to evaluation and deployment.



Using LLMs to Build ML Models

How large language models support each stage of the ML pipelline

LLM

Always-on support across the entire ML pipeline

1. **Problem Framing**
   Turn business or clinical needs into ML tasks
2. **Data Collection**
   Define data needs, labeling rules, and queries
3. **Data Preparation**
   Suggest cleaning, imputation, and feature ideas
4. **Model Selection**
   Recommend suitable algorithms and baselines
5. **Training & Tuning**
   Generate starter code and hyperparameter ideas
6. **Evaluation**
   Explain metrics, errors, bias, and validation
7. **Deployment & Monitoring**
   Help with APIs, documentation, alerts, and retraining

LLMs accelerate ideation, coding, explanation, and documentation — while humans still provide judgment, domain knowledge, and final decisions.

Hands on using LLM and Google Colab to build ML model using NHIS Healthcare Claims and Fraud Dataset

Practical workshop for healthcare claims analytics and fraud detection

DATA PEOPLE AI BETTER HEALTHCARE

1 NHIS Claims & Fraud Dataset

claim_id	patient_id	provider_id	amount	fraud_label
CLM001	P001	H001	12,340	0
CLM002	P002	H158	56,870	1
CLM003	P003	H026	8,920	0
...	...	...	...	...

Real-world NHIS healthcare claims and fraud dataset

2 Data Preparation

- Data cleaning
- Feature engineering
- Handle missing values
- Prepare training data

Clean, transform and create meaningful features

3 LLM Support

- Write and debug code
- Explain results
- Suggest features
- Generate visualizations

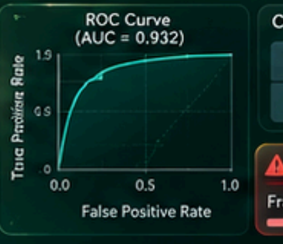
Prompt engineering, code assistance and explanation

4 Google Colab Model Building

```
model = RandomForestClassifier()
model.fit(X_train, y_train)
pred = model.predict(X_test)
```

- Train machine learning model
- Tune parameters
- Use GPU (Colab)
- Analyze feature importance

5 Fraud Prediction & Evaluation



Confusion Matrix

8912	189
247	652

High Fraud Risk Claim
Fraud probability: 0.87

Predict fraud, evaluate performance and derive actionable insights

From healthcare claims data to interpretable fraud detection workflows

REAL DATA | REAL INSIGHTS | SAFER HEALTHCARE

Vibe Coding for Health Apps

Turn your ideas into working apps with natural language

Describe your idea → Generate code → Preview and test → Refine with prompts → Deploy your app

Google AI Studio

- Build full-stack apps with Gemini
- Multimodal (text, image, data)
- Good for rapid prototyping

Chat | Build | Preview

Create a health symptom checker web app...

Claude Code

- Generate and edit full apps
- Clean, well-structured code
- Great for iterative development

Chat | Code | Preview

Build a patient appointment app with calendar and notification...

OpenAI ChatGPT

- Generate, refactor and explain code
- Create UI, APIs and database
- Good for quick prototypes and ideation

Chat | Code | Preview

Create a modern health app with user login and symptom checker...

Cursor

- AI-first code editor
- Understands and edits large codebases
- Fast for building and iterating

Replit

- Build, run and deploy in browser
- Collaborative development
- Quick sharing and testing

Lovable

- Turn ideas into full apps with natural language
- Beautiful UI out of the box
- Great for non-developers

Create a health tracking app with modern design...

Hands on using Google AI Studio to build Apps using NHIS Healthcare Claims and Fraud Dataset

Practical workshop for healthcare claims analytics, fraud detection, and app prototyping

1 NHIS Claims & Fraud Dataset

claim_id	patient_id	provider_id	amount	fraud_label
CLM001	P1001	H001	12,340	0
CLM002	P2320	C045	56,780	1
CLM003	P3188	H210	8,920	0

Healthcare claims records, provider data, patient data, fraud labels, and transaction patterns

2 Data Preparation

- Clean data
- Handle missing values
- Explore fraud indicators
- Prepare structured inputs

Clean data, handle missing values, explore fraud indicators, and prepare structured inputs

3 Google AI Studio

- Design app logic
- Generate interfaces
- Create workflows
- Refine use cases

Use prompts to design app logic, create interfaces, generate workflows, and refine use cases

4 Build Apps

- Create claims dashboards
- Fraud screening tools
- Reporting forms
- Case review apps
- Assistant-style workflows

Create claims dashboards, fraud screening tools, reporting forms, case review apps, and assistant-style workflows

5 Testing & Insights

Fraud Aert Review

- Test features
- Review fraud alerts
- Analyze results
- Improve experience

Test features, review fraud alerts, analyze results, and improve the application experience

From healthcare claims data to practical AI-powered applications

REAL DATA | PRACTICAL APPS | TRUSTED AI | A HEALTHIER TOMORROW

THANK YOU

The future of health is not AI replacing humans. it is humans empowered by AI to make better decisions, earlier.